Exact non-scattering homogeneous dielectrics from Schiffer counterexamples

Two-page technical summary for Matthew Colbrook and George Stepaniants

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Claim. Using your authenticated regular Schiffer zero as the sole large computer-assisted input, the same coefficient ball appears to contain a family of exact positive-frequency transmission solutions. After scaling, the exterior wavenumber is fixed at $k = 1$ and the incident field is the normalized Herglotz wave

$$v(x) = \sqrt{2\pi}J_0(|x|), \qquad h(\theta) = 1/\sqrt{2\pi}.$$

For every $0 < \tau \leq 10^{-13}$ the proposed exact domain and contrast are

$$\Omega_\tau = \sqrt{\tau}\,\phi_{p_\tau}(\mathbb{D}), \qquad q_\tau = p_{\tau,0}^2/\tau.$$

The domain stays in your certified univalent non-disc ball, $q_\tau > 0$ and $q_\tau \neq 1$, and both physical traces match exactly.

Perturbed fixed-disc equation. Put

$$H_{\tau,p} = J_0(\sqrt{\tau}|\phi_p|) = \sum_{n\geq 0} \frac{(-1)^n}{(n!)^2}\left(\frac{\tau|\phi_p|^2}{4}\right)^n.$$

Conformal covariance gives

$$\Delta H_{\tau,p} + \frac{\tau}{p_0^2}|p|^2 H_{\tau,p} = 0.$$

The exact equation is therefore

$$F_\tau(\gamma, p) = \gamma + |p|^2 K\gamma + |p|^2\left(1 - \frac{\tau}{p_0^2}\right)H_{\tau,p} = 0.$$

It reduces literally to your Schiffer equation at $\tau = 0$. If $W = K\gamma$, then $H_{\tau,p} + W$ solves the pulled-back interior equation, and $W = \partial_r W = 0$ on the unit circle supplies exact Cauchy matching.

Conservative quantitative closure. Only the simple authenticated majorants

$$Y \leq 1.59\times 10^{-10}, \quad Z \leq 0.621, \quad C_2 \leq 122, \quad C_3 \leq 0.012, \quad r = 10^{-6}$$

are imported. Since your formula is $C_3 = \alpha/(96b^2)$ and the exact center has $b < 32$,

$$\alpha < 0.012 \cdot 96 \cdot 32^2 = 1179.648 < 1180.$$

On the whole ball, $\|p\| < 55$, $p_0 > 31$, and $\phi = z\psi$ with $\|\psi\| < 1.1$, hence $\||\phi|^2\| < 5/4$. The complete Bessel series, without truncation, gives

$$\|F_\tau - F_0\| < 1906\tau, \qquad \|DF_\tau - DF_0\| < 3\tau.$$

Consequently, at the endpoint $\tau = 10^{-13}$,

$$R_\tau \leq -1.53810999999988 \times 10^{-7} < 0, \qquad L_\tau \leq 0.621244000354036 < 1.$$

Monotonicity gives the entire interval. The regular-zero interpretation also follows abstractly from your invertible preconditioner and the Neumann bound, so the Banach-space implicit-function theorem produces a local branch before any quantitative endpoint is chosen.

Physical scaling. In the unscaled domain, the exterior squared wavenumber is τ and the interior squared wavenumber is $p_{\tau,0}^2$. With $y = \sqrt{\tau}x$, the exterior equation becomes $(\Delta_y + 1)v = 0$ and the interior equation becomes

$$(\Delta_y + p_{\tau,0}^2/\tau)U = 0.$$

The zero clamped traces remain zero under conformal and dilation factors. Thus the exterior total field is exactly v and the scattered field is the zero radiating solution.

What I would most value you checking.

1. Is the operator F_τ expressed in exactly the same fixed-disc normalization and compatible range as your F_0?

2. Is the use of $C_3 = \alpha/(96b^2)$ and the imported shape norm convention exactly faithful to the released certificate?
3. Do you agree that the contraction proof makes the exact Schiffer zero regular via invertibility of $\mathcal{A}DF_0 = I - DT_0$?
4. Is there any attribution, collaboration, or dependency language you would prefer before a public preprint is posted?

Formal and executable checks. The supplement includes exact-rational Python and Node verifiers that rederive the perturbation constants from your certified ball bounds, a replay of your authenticated checker from the archived Zenodo release with a machine comparison of every imported value, and a narrow Lean 4/mathlib re-verification of the rational certificate arithmetic together with generic fixed-point lemmas. The release explicitly imports, rather than reclaims, your MPFR theorem; the standard Helmholtz/Bessel analysis is proved in the article, not in Lean. A fresh pinned-toolchain build is retained as a pre-publication gate before any Lean source is described as kernel-checked.

Attached material. The focused research article, the verification supplement, this note, the exact certificate, the Lean project, and the one-command reproduction archive.